NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83401/13
13 August 1991

PERFORMANCE SPECIFICATION

RESISTOR NETWORK, FIXED, FILM,
STYLE RZ130, 14 PIN DIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RZ130, 14 pin dual-in-line package (DIP) resistor networks. This style is available in all characteristics and resistance tolerances.

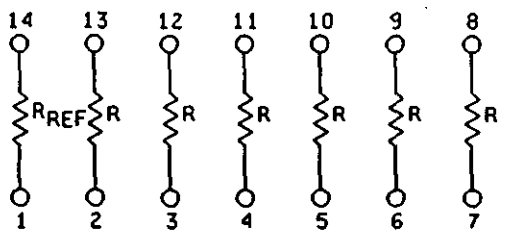
1.2 Military part number. Resistor networks covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:

M8340113 — Detail specification number

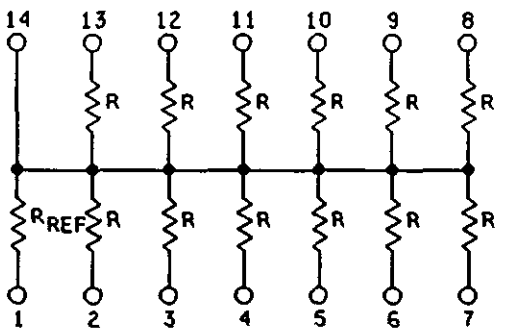
H1002FA — Coded number

The coded number shall be derived in accordance with MIL-R-83401.

1.2.1 Schematics. The schematic of the resistor network shall be identified by a single letter in accordance with the drawings below. The resistor element R_{REF} shall be the reference element used in determining the ratio accuracy (when applicable).



SCHEMATIC A



SCHEMATIC B

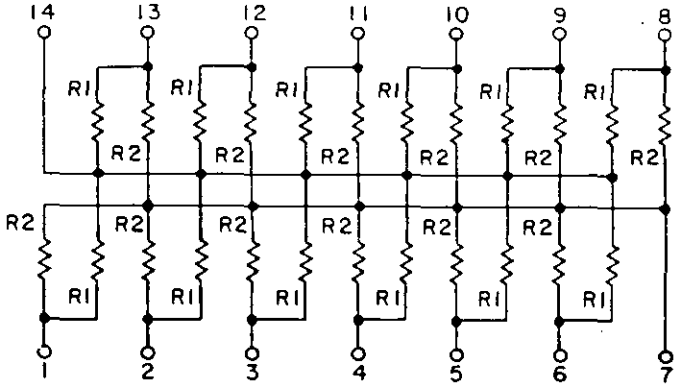
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5905

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SCHEMATIC J

1.2.1.1 Standard resistance values. The standard resistance values and the resistance value designators for the J schematic are as specified in table I.

TABLE I. Standard resistance values.

Resistance designator	R_1 (ohms)	R_2 (ohms)	Resistance designator	R_1 (ohms)	R_2 (ohms)
A001	82	130	A010	330	470
A002	120	200	A011	330	680
A003	130	210	A012	1.5k	3.3k
A004	160	260	A013	3.0k	6.2k
A005	180	240	A014	180	270
A006	180	390	A015	270	270
A007	220	270	A016	560	560
A008	220	330	A017	560	1.2k
A009	330	390	A018	620	2.7k

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-39032 - Resistors, Packaging of.
MIL-R-83401 - Resistor Networks, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Resistors, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-83401, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Internal connection. Internal connections shall be accomplished by thermal compression bonding of gold to gold, to form a solid phase weld from the lead to the internal resistor connection. No solder connection, eutetic or noneutetic, shall be permitted in this construction.

3.3 Power rating (70°C). The power ratings for schematics A, B, and J shall be as specified in table II.

TABLE II. Power ratings.

Schematic	Characteristic	Element power rating (watts)	Network power rating (watts)
A	C	0.1	0.7
	V	0.1	0.7
	H	0.2	1.4
	K	0.2	1.4
	M	0.2	1.4
B	C	0.05	0.65
	V	0.05	0.65
	H	0.1	1.3
	K	0.1	1.3
	M	0.1	1.3
J	C	0.025	0.6
	V	0.025	0.6
	H	0.050	1.2
	K	0.050	1.2
	M	0.050	1.2

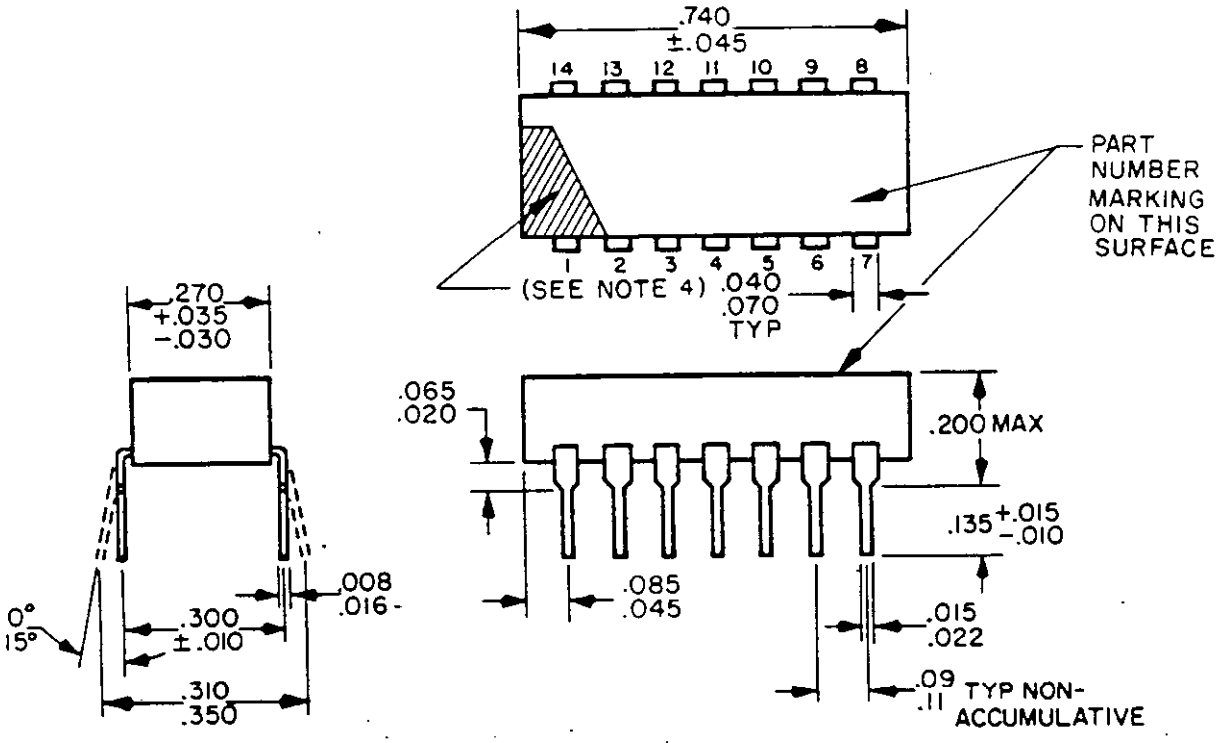
3.4 Power conditioning. The power applied for power conditioning for schematics A, B, and J shall be 1.5 times rated power.

3.5 Voltage rating. The maximum continuous working voltage for each resistor shall not exceed 100 volts dc or ac rms.

3.6 Resistance. Minimum and maximum resistance values shall be as follows:

Resistance tolerance	Minimum resistance (ohms)	Maximum resistance (megohm)
B (±0.1 percent)	100	0.1
D (±0.5 percent)	100	0.1
F (±1.0 percent)	10.0	1
G (±2.0 percent)	10.0	1
J (±5.0 percent)	10.0	1

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Inches	mm	Inches	mm
.008	0.20	.070	1.78
.010	0.25	.085	2.16
.015	0.38	.090	2.29
.016	0.41	.110	2.79
.020	0.51	.135	3.43
.022	0.56	.200	5.08
.030	0.76	.270	6.86
.035	0.89	.300	7.62
.040	1.02	.310	7.87
.045	1.14	.350	8.89
.065	1.65	.740	18.80

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The picturization of the style above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.
4. Pin 1 locator shall be a dot, notch, stripe, or numeral 1 adjacent to pin no. 1, in the shaded area.

FIGURE 1. Style RZ130 resistors.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-83401.

4.2 J schematic tests. For the tests on J schematic, the dc resistance measurements, resistance temperature characteristic measurement, and short time overload power application shall be made between each terminal and the applicable common terminal without compensation for the shunt circuits. The pin to pin dc resistance measurements shall be used directly to determine the acceptability to section 3 requirements.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-R-83401 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5905-1270-1)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-83401/13	2. DOCUMENT DATE *(YYMMDD)*

3. DOCUMENT TITLE
Resistor Network, Fixed, Film, Style RZ130, 14 Pin DIP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE** *(Include Area Code)* **(1) Commercial** **(2) AUTOVON** *(If applicable)*	**7. DATE SUBMITTED** *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 98-290

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